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SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

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Note

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METHOD AS TOOL - INTERAZIONE CON DB

Note

[illegible]

- Simone Scannapieco

[illegible]

```
...
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>
<dependency>
  <groupId>org.projectlombok</groupId>
  <artifactId>lombok</artifactId>
  <scope>annotationProcessor</scope>
</dependency>
...
```

```
spring:
  ...
  jpa:
    hibernate:
      ddl-auto: update
```

[illegible]

Sei un assistente virtuale di helpdesk, responsabile di assistere i clienti con i loro problemi. Il tuo obiettivo principale è fornire soluzioni chiare, accurate e utili ai problemi più comuni. Se il problema di un utente non può essere risolto tramite la tua risposta, offriti di:

- Creare un nuovo ticket helpdesk per il problema segnalato dal cliente
- Verificare lo stato dei ticket helpdesk esistenti

Cerca sempre di essere cortese, proattivo e orientato alle soluzioni.

Aderisci alle seguenti linee guida:

- Se esiste già un ticket relativo allo stesso problema per lo stesso cliente, non creare duplicati
- Se non sei sicuro della natura del problema prima di creare il ticket, chiedi al cliente di spiegare meglio la problematica o chiedi maggiori informazioni
- Chiedi sempre conferma esplicita al cliente prima di aprire un nuovo ticket
- Quando rispondi riguardo allo stato dei problemi esistenti, mantieni la risposta breve con lo stato del ticket e il tempo stimato di risoluzione (ETA).
- Usa i tool forniti dall'utente per fornire la tua risposta

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```
package it.venis.ai.spring.demo.entity;

import java.time.LocalDateTime;

import jakarta.persistence.*;
import lombok.*;

@Entity
@Getter
@Setter
@Builder
@NoArgsConstructor
@AllArgsConstructor
@Table(name = "helpdesk_tickets")
public class HelpDeskTicket {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String username;

    private String issue;

    private String status; // e.g., OPEN, IN_PROGRESS, CLOSED

    private LocalDateTime createdAt;

    private LocalDateTime eta;
}
```

[illegible]

```
package it.venis.ai.spring.demo.model;

public record TicketRequest(String issue) {
}
```

Note

[illegible]


```
package it.venis.ai.spring.demo.repository;

import java.util.List;

import org.springframework.data.jpa.repository.JpaRepository;

import it.venis.ai.spring.demo.entity.HelpDeskTicket;

public interface HelpDeskTicketRepository extends JpaRepository<HelpDeskTicket, Long> {

    List<HelpDeskTicket> findByUsername(String username);

}
```

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```
package it.venis.ai.spring.demo.services;

...

@Service
@RequiredArgsConstructor
public class HelpDeskTicketService {

    private final HelpDeskTicketRepository helpDeskTicketRepository;

    public HelpDeskTicket createTicket(TicketRequest ticketInput, String username) {

        HelpDeskTicket ticket = HelpDeskTicket.builder()
            .issue(ticketInput.issue())
            .username(username)
            .status("OPEN")
            .createdAt(LocalDateTime.now())
            .eta(LocalDateTime.now().plusDays(7))
            .build();

        return helpDeskTicketRepository.save(ticket);

    }

    public List<HelpDeskTicket> getTicketsByUsername(String username) {

        return helpDeskTicketRepository.findByUsername(username);

    }

}
```

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

Help desk tools - I

```
package it.venia.ai.spring.demo.tools;

...

@Component
@RequiredArgsConstructor
public class HelpDeskTools {

    private static final Logger LOGGER = LoggerFactory.getLogger(HelpDeskTools.class);

    private final HelpDeskTicketService service;

    @Tool(name = "createTicket",
        description = "Genera un nuovo ticket di assistenza.",
        returnDirect = true)
    String createTicket(
        @ToolParam(description = "Il problema segnalato dall'utente.") TicketRequest ticketRequest,
        ToolContext toolContext) {

        String username = (String) toolContext.getContext().get("username");
        LOGGER.info("Creazione ticket di assistenza per utente {} con dettagli {}", username, ticketRequest);
        HelpDeskTicket savedTicket = service.createTicket(ticketRequest, username);
        LOGGER.info("Ticket creato con successo. Ticket ID: {}, Username: {}",
            savedTicket.getId(),
            savedTicket.getUsername()
        );
        return "Ticket #" + savedTicket.getId() + " creato con successo per utente " + savedTicket.getUsername();
    }

    ...
}
```

Note

[illegible]

```
...
@Tool(description = "Recupera lo stato dei ticket di assistenza a partire da uno specifico username.")
List<HelpDeskTicket> getTicketStatus(ToolContext toolContext) {

    String username = (String) toolContext.getContext().get("username");
    LOGGER.info("Recupero ticket per utente: {}", username);
    List<HelpDeskTicket> tickets = service.getTicketsByUsername(username);
    LOGGER.info("Trovati {} ticket di assistenza per utente {}", tickets.size(), username);
    return tickets;
}
}
```

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```
package it.venis.ai.spring.demo.config;

...

@Configuration
public class HelpDeskChatClientConfig {

    @Value("classpath:/templates/get-help-desk-system-prompt.st")
    Resource helpDeskSystemPrompt;

    @Bean
    public ChatClient geminiHelpDeskToolsChatClient(
        OpenAiChatModel geminiChatModel,
        HelpDeskTools helpDeskTools,
        ChatMemory chatMemory,
        @Qualifier("messageChatMemoryAdvisor") BaseChatMemoryAdvisor chatMemoryAdvisor) {

        ChatClient.Builder chatClientBuilder = ChatClient.builder(geminiChatModel);

        return chatClientBuilder
            .defaultTools(helpDeskTools)
            .defaultAdvisors(List.of(new SimpleLoggerAdvisor(), chatMemoryAdvisor))
            .defaultSystem(helpDeskSystemPrompt)
            .build();
    }

    ...
}
```

Note

[illegible]

Configurazione ChatClient - II

```

...
@Bean
public ChatClient ollamaHelpDeskToolsChatClient(
    OllamaChatModel ollamaChatModel,
    HelpDeskTools helpDeskTools,
    ChatMemory chatMemory,
    @Qualifier("messageChatMemoryAdvisor") BaseChatMemoryAdvisor chatMemoryAdvisor) {

    ChatClient.Builder chatClientBuilder = ChatClient.builder(ollamaChatModel);

    return chatClientBuilder
        .defaultTools(helpDeskTools)
        .defaultAdvisors(List.of(new SimpleLoggerAdvisor(),
                                new OllamaCostSavingsAdvisor(),
                                chatMemoryAdvisor))
        .defaultSystem(helpDeskSystemPrompt)
        .build();
}
}

```

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```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@Configuration
public class QuestionController {

    ...

    private final ChatClient geminiHelpDeskToolsChatClient;
    private final ChatClient ollamaHelpDeskToolsChatClient;

    public QuestionController(QuestionService service,
        RAGService ragService,
        TimeToolsService timeToolsService,
        @Qualifier("geminiWeatherToolsChatClient") ChatClient geminiWeatherToolsChatClient,
        @Qualifier("ollamaWeatherToolsChatClient") ChatClient ollamaWeatherToolsChatClient,
        @Qualifier("geminiHelpDeskToolsChatClient") ChatClient geminiHelpDeskToolsChatClient,
        @Qualifier("ollamaHelpDeskToolsChatClient") ChatClient ollamaHelpDeskToolsChatClient) {

        this.service = service;
        this.ragService = ragService;
        this.timeToolsService = timeToolsService;
        this.geminiWeatherToolsChatClient = geminiWeatherToolsChatClient;
        this.ollamaWeatherToolsChatClient = ollamaWeatherToolsChatClient;
        this.geminiHelpDeskToolsChatClient = geminiHelpDeskToolsChatClient;
        this.ollamaHelpDeskToolsChatClient = ollamaHelpDeskToolsChatClient;

    }
}
```

Note

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```
...
@PostMapping("/gemini/ask/help-desk-tools/help-desk")
public Answer getGeminiHelpDeskToolAnswer(@RequestHeader("username") String username,
    @RequestParam("message") String message) {
    return new Answer(
        this.geminiHelpDeskToolsChatClient.prompt()
            .advisors(a -> a.param(ChatMemory.CONVERSATION_ID, username))
            .user(message)
            .toolContext(Map.of("username", username))
            .call().content()
    );
}

@PostMapping("/ollama/ask/help-desk-tools/help-desk")
public Answer getOllamaHelpDeskToolAnswer(@RequestHeader("username") String username,
    @RequestParam("message") String message) {
    return new Answer(
        this.ollamaHelpDeskToolsChatClient.prompt()
            .advisors(a -> a.param(ChatMemory.CONVERSATION_ID, username))
            .user(message)
            .toolContext(Map.of("username", username))
            .call().content()
    );
}
}
```

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Branch: 14-spring-ai-gemini-ollama-help-desk-tools

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